

Day 3 Gap Failure Screener (ATR)

What the Pine Script does, how to use it, its inputs and outputs, and what to expect in TradingView Pine Screener.

Primary file day3_gap_failure_pine_screener.pine	Script type Indicator for Pine Screener	Best use 5-minute US equities short setup scanner
Logic theme Three-day momentum followed by day-3 failure	ATR role Reference metric and target output	Orders placed None in screener version

Executive summary

This script is a screener-oriented rewrite of your earlier ATR-based short setup. It does not enter or manage trades. Instead, it scans symbols for a very specific sequence: a strong Day 1 gap, Day 2 follow-through, and then a Day 3 gap or opening drive that fails. Once that failure happens, the script waits for a confirmed lower-high pivot before flagging a short signal.

Signal lifecycle used by the Pine Screener script

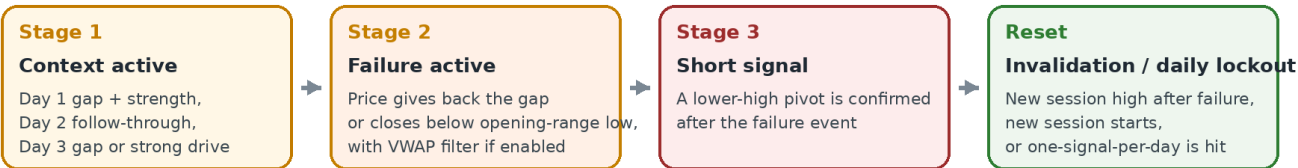


Figure 1. High-level signal lifecycle used by the screener.

1. Purpose and scope

This Pine Script is designed to find potential intraday short candidates after a strong two-day momentum run. The idea is that some stocks gap and trend strongly for two sessions, then attempt to extend again on Day 3. When that Day-3 continuation fails, the script looks for a lower-high confirmation and surfaces the symbol as a short setup.

Because you asked for a TradingView screener version, the file is built as an indicator rather than a strategy. That distinction matters: the script emits screening columns and alerts, but it does not place entries, stops, or exits.

Topic	Explanation
Primary purpose	Scan a watchlist for a Day-3 gap-or-drive failure short pattern.
Best environment	Regular-session US equities on a 5-minute timeframe.
Earliest useful time	After the opening range is complete. With the default settings on a 5-minute chart, that is after the first 30 minutes.
What it does not do	It does not backtest P&L, place simulated orders, or manage trade exits in the screener version.
User-supplied inputs required	None. Every input has a default value. You only change inputs when you want to tighten, loosen, or customize the scan.
Core output	A binary short signal, a failure-active state, a context-active state, ranking metrics, and two alert conditions.

Required vs. optional

Required from the user: nothing. The script runs with defaults. Required from the environment: intraday data, daily bars for context, and a 5-minute timeframe if `strict5m` is left on. Everything else is optional tuning.

2. What the script does, step by step

1. It checks whether the script is running during the defined trading session and, if `strict5m` is enabled, whether the chart or screener timeframe is exactly 5 minutes.
2. It tracks the current session open, high, low, and opening range. By default, the opening range uses 6 five-minute bars, which equals 30 minutes.
3. It pulls higher-timeframe daily context from prior completed daily bars: the previous close, previous high, daily ATR, the Day-1 gap percentage, whether Day 1 was strong, and whether Day 2 showed follow-through.
4. It measures whether Day 3 qualifies as an extension attempt by checking either a minimum opening gap or a strong opening drive, optionally including a break of the prior day's high.

5. If the Day-3 context is valid, it watches for failure. Failure means the stock gives back the gap by closing below the previous close, or it closes below the opening-range low. If the VWAP filter is enabled, price also needs to be below VWAP at the failure point.
6. After failure is recorded, the script waits for a lower-high pivot to be confirmed. This confirmation uses `ta.pivohigh(high, pivotLeft, pivotRight)`, so the signal appears only after the right-side confirmation bars have closed.
7. Once the lower-high pivot is confirmed, the script sets Short Signal to 1 on that bar, resets the failure state, and—if `oneSignalPerDay` is enabled—prevents additional same-day signals for that symbol.

Stage	Name	Meaning	How to interpret it
0	Inactive	The symbol does not currently meet the full setup requirements.	Ignore it or wait.
1	Context active	The stock has the higher-timeframe setup and a valid Day-3 extension context.	Put it on watch, but there is no failure yet.
2	Failure active	The setup failed intraday and is waiting for a lower-high confirmation.	This is the stalking stage.
3	Short signal	A lower-high pivot was confirmed after the failure event.	This is the current-bar entry candidate output.

3. Inputs and how they affect outcomes

All inputs are optional because the script provides defaults. The practical question is not whether an input is required, but whether changing it makes the screener more selective, faster, slower, cleaner, or noisier.

Setup input	Default	Purpose	Impact on outcomes
tradeSession	0930-1600	Defines when the intraday logic is active.	Changing the session changes what bars are included in the setup and can materially alter signals.
strict5m	true	Requires the script to run only on a 5-minute timeframe.	When true, accidental use on other timeframes produces inactive outputs rather than distorted logic.
openingRangeBars	6	Number of bars used for the opening range.	Higher values delay evaluation and make the opening range wider; lower values make the script react earlier.
minDay1GapPct	3.0	Minimum Day-1 opening gap percentage.	Higher values restrict the scan to stronger momentum names and reduce signal count.

minDay2CloseGainPct	0.5	Minimum Day-2 close-over-close follow-through.	Higher values demand more sustained continuation into Day 2.
minDay3GapPct	0.5	Minimum Day-3 gap percentage.	Higher values require a more obvious Day-3 extension attempt.
minDrivePct	1.0	Minimum opening-drive percentage if there is no Day-3 gap.	Higher values reduce false positives from small, weak early pushes.
usePrevDayHighBreak	true	Lets an opening-range break above the prior day high count as a strong drive.	When on, more charts can qualify even without a large pure percentage drive.
pivotLeft	2	Bars to the left of the pivot for lower-high confirmation.	Larger values make the pivot structurally stronger but slower.
pivotRight	2	Bars to the right of the pivot for confirmation.	Larger values delay the signal but can reduce noise.
minPrice	1.0	Minimum stock price allowed.	Raising it helps exclude very low-priced names and can improve screen quality.
oneSignalPerDay	true	Allows only one signal per symbol per day.	Turning it off can create multiple same-day signals after resets.

Filter / view input	Default	Purpose	Impact on outcomes
atrLen	14	Length used for daily ATR.	Higher values smooth ATR more; lower values make ATR more reactive.
targetAtr	0.5	ATR multiple used only to calculate the output column ATR Target \$.	Important: this does not place an exit order in the screener version.
requireBelowVWAP	true	Requires price to be below VWAP at failure and entry.	When on, the script is stricter and usually produces fewer but cleaner countertrend-failure setups.
showChartBackground	false	Optional chart shading based on setup stage.	Purely visual. It does not affect the screener logic or outputs.

Most important tuning levers

The inputs that change behavior the most are openingRangeBars, minDay1GapPct, minDay2CloseGainPct, minDay3GapPct, minDrivePct, pivotLeft, pivotRight, and requireBelowVWAP.

Together they determine how selective the scanner is, how quickly it reacts, and how much confirmation it demands.

4. Outputs and what to expect

The screener version exposes its results through hidden plot outputs and alert conditions. These outputs are meant to become columns and filters inside Pine Screener.

Output	Type	Meaning	Typical use
Short Signal	0 or 1	1 on the bar where the lower-high confirmation produces a short candidate.	Use it as the main entry filter.
Failure Active	0 or 1	1 after a failure event has been detected and before a signal or invalidation occurs.	Use it to find setups that are close but not yet confirmed.
Context Active	0 or 1	1 when the full three-day context exists, even if there is no failure yet.	Use it as an early watchlist stage.
Setup Stage	0, 1, 2, 3	The state machine value described earlier.	Excellent as a sort or grouping column.
Setup Score	0–100 heuristic	A weighted ranking score based on setup components.	Helpful for sorting; it does not gate signals.
Day1 Gap %	Percent	Opening gap percentage of the first day in the three-day sequence.	Useful for judging momentum intensity.
Today Gap %	Percent	Current Day-3 opening gap versus the prior close.	Useful for checking extension quality.
Opening Drive %	Percent	Measured opening move from session open to opening-range high.	Useful when there is no large Day-3 gap.
Daily ATR	Price units	Prior completed daily ATR.	Reference only in the screener version.
ATR Target \$	Price units	Daily ATR multiplied by targetAtr.	Reference target distance only, not an order.

The script also defines two alert conditions: Day 3 Short Entry and Day 3 Failure Active. These correspond to the confirmed signal and the active-failure stalking state.

Very important interpretation note

Setup Score does not generate signals by itself. It is only a ranking layer. A high score can be useful when sorting watchlist names, but Short Signal, Failure Active, and Setup Stage are the operational outputs.

5. When to run it

This script is most useful during the regular session, especially after the opening range has completed. With the default settings on a 5-minute chart, the opening range finishes after the first six bars, so the first fully eligible bar is the bar after 10:00 a.m. Eastern.

Time window	What to expect
Before the open	Prepare your watchlist, but expect no valid outputs yet because intraday session variables have not formed.
9:30–10:00 ET	The script is still building the opening range. Context may exist in daily data, but failure logic is not fully actionable yet.
After 10:00 ET	This is the main operating window. Stage 1, Stage 2, and Stage 3 become meaningful.
Midday onward	Signals can still appear, but the pattern is fundamentally an opening extension/failure idea, so quality often depends on how the morning developed.
After the session	Outputs reset with the next session. This is not designed for after-hours scanning.

8. Open Pine Screener on a watchlist of liquid stocks.
9. Set the screener timeframe to 5 minutes.
10. Use Setup Stage or Context Active to find names entering the setup.
11. Use Failure Active to locate names that have failed but are still awaiting confirmation.
12. Use Short Signal = 1 to isolate symbols where the lower-high confirmation occurred on the current bar.
13. Review the chart manually for liquidity, spread, news, volume, and your own execution rules.

6. Key variables inside the script

Variable / series	Why it matters
tfOkay	True only when the timeframe passes the strict5m check, or when strict5m is disabled.
inSession / newSession / endSession	Control whether the intraday logic is currently active and when daily state variables reset.
barsFromOpen	Counts bars from the session open so the script knows when the opening range has completed.
sessOpen / sessHigh / sessLow	Track the current session open and running high/low values.
orHigh / orLow	Store the opening-range high and low used for drive and failure logic.
failureSeen / failureBarIndex	Persist the failure state and remember where it began.
signaledToday	Prevents repeated signals when oneSignalPerDay is enabled.

prevClose / prevHigh	Pull prior completed daily reference points used for Day-3 gap and drive tests.
dailyATR	Prior completed daily ATR used as a reference metric.
day1GapPct / day1Strong / day2FollowThrough	Define whether the first two days show the required momentum backdrop.
todayGapPct / openingDrivePct	Measure the current day's extension attempt.
hasDay3Gap / hasStrongDrive	Determine whether the Day-3 extension condition is satisfied.
threeDayContext	The master context flag: strong Day 1 + Day 2 follow-through + valid Day-3 extension attempt.
gaveBackGap / engulfedOpeningDrive	The two main failure conditions.
pivotHigh / lowerHighConfirmed	The pivot-based confirmation logic used to convert failure into a short signal.
stage	The compact state-machine output used for screening and sorting.
setupScore	The script's weighted ranking number, not a trading instruction.
atrTargetDollars	Computed as dailyATR multiplied by targetAtr; informational only in this version.

7. How input changes affect results

The fastest way to change the character of the screener is to decide whether you want earlier, more frequent signals or later, cleaner signals.

- More selective scanner: raise minDay1GapPct, minDay2CloseGainPct, minDay3GapPct, and minDrivePct; keep requireBelowVWAP on; consider raising minPrice.
- Earlier but noisier confirmation: reduce pivotLeft and pivotRight. This can surface signals sooner, but it also increases the chance of noise.
- Slower but cleaner confirmation: increase pivotLeft and pivotRight. You will miss some early entries, but the pivot structure becomes more demanding.
- Wider operating universe: lower minPrice or loosen gap thresholds. This increases coverage, but it can also introduce lower-quality names.
- Tighter opening behavior: raise openingRangeBars if you want the morning range to be defined over more bars. This delays eligibility but can reduce whipsaw behavior.

Profile	Suggested configuration
Selective / institutional feel	Keep strict5m=true, requireBelowVWAP=true, openingRangeBars=6, minDay1GapPct around 3–5, minDay2CloseGainPct around 0.5–1.0, and pivotLeft/pivotRight at 2 or higher.
Aggressive / earlier alerts	Consider minDrivePct below 1.0 and pivotLeft/pivotRight at 1, but expect more noise and more manual chart review.
Cleaner universe	Raise minPrice and use your own watchlist filters for liquidity, spread, and relative volume.

8. Limitations, assumptions, and gotchas

- This is a screener indicator, not an execution engine. It does not submit orders or calculate strategy performance.
- ATR is informational here. The screener script outputs ATR Target \$, but it does not place target or stop orders.
- Pivot confirmation is delayed by pivotRight bars. That delay is intentional, because the script waits for confirmation before signaling.
- With strict5m enabled, using another timeframe will intentionally suppress signals.
- The openingRangeBars default assumes a 5-minute chart. If you disable strict5m and use another timeframe, 6 bars no longer means 30 minutes.
- There is no built-in volume, float, news, borrow availability, sector, or market-regime filter.
- Failure can be invalidated if a new session high is made after the failure event and before confirmation.
- The script resets state at a new session. It is built for regular-session intraday behavior, not after-hours behavior.
- Daily context uses completed daily bars and lookahead is disabled, which helps avoid future leakage from the higher timeframe.

Practical best practice

Treat the script as a candidate finder, not a complete trading plan. The best workflow is: let the screener narrow your watchlist, then confirm liquidity, relative volume, trend context, news risk, and your own execution rules on the chart.

9. Relationship to the earlier strategy version

Feature	Screener version	Strategy version
Type	Indicator for Pine Screener	Backtestable strategy
Main job	Surface candidates	Simulate entries and exits
Short entry	Signals only	Places a strategy.short entry
ATR use	Reference outputs only	Uses ATR for target and stop buffers
VWAP use	Filter for failure and entry	Filter plus exit-on-VWAP-reclaim option
Best use case	Scanning and ranking symbols	Studying execution and trade management

10. Formula summary

A few formulas are especially helpful when you want to understand what the columns mean:

- Day1 Gap % = $100 \times (\text{open}[2] - \text{close}[3]) / \text{close}[3]$
- Today Gap % = $100 \times (\text{session open} - \text{previous close}) / \text{previous close}$
- Opening Drive % = $100 \times (\text{opening-range high} - \text{session open}) / \text{session open}$

- $\text{ATR Target \$} = \text{daily ATR} \times \text{targetAtr}$

The script also computes a heuristic Setup Score by awarding weighted points to key setup components. That score is useful for sorting, but it is not part of the final trigger condition.

11. Bottom line

In plain English, this script watches for a stock that had a strong Day 1, followed through on Day 2, tried to extend again on Day 3, then failed. When that failure is confirmed by a lower high, the screener marks it as a short candidate. Its strength is in narrowing a watchlist quickly; its limitation is that you still need chart judgment and execution rules.

Suggested first use

Run it on a liquid watchlist at 5-minute resolution after the first 30 minutes of the session. Sort by Setup Stage, then by Setup Score. Use Failure Active to stalk names, and use Short Signal for current-bar triggers.